

Data Quality Analysis Report

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Executive Summary

This report analyzes data quality across 1 files containing 1,691 total records. The analysis identified 9,864 validation errors. **Key Recommendations:** • Address mandatory field completeness issues at the source • Implement data validation workflows before processing • Review critical validation errors for immediate remediation

File Quality Scores

File Name	Mandatory Complete	Total Records	Errors	Quality %
Activity_Curriculum.xlsx	100.0%	1,691	9,864	42.5%

Error Summary

Error Type	Count	Severity	Recommended Action
ENCODING_ISSUE	3,804	High	Source: invalid characters. Fix character encoding.
MANDATORY_EMPTY	3,382	Critical	Source: mandatory field empty. Requires data correction.
HETEROGENEOUS_TYPE	1,288	High	Workshop: mixed data types. Standardize approach needed.
LEADING_SPACES	1,158	Medium	Transform: leading whitespace. Auto-fix with trim.
TRUNCATION	232	High	Source: field exceeds length limit. Adjust target or truncate data.

Cross-File Referential Integrity Analysis

This section analyzes referential integrity between files using dynamic relationship discovery from Neo4j field mappings. Critical integrity issues can impact data migration success.

Activities Domain Integrity Analysis

Business rule-based orphan detection for Activities domain relationships. This analysis identifies ILT courses and sessions that lack valid activity references.

Medium Priority Issues (3 relationships)

Relationship	Integrity %	Orphaned Records
Activities.ClassCode → Activities.Activity Code (report-only)	0.0%	451.0

Activities.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	451.0
Activities.SessionParts.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	1.0

Integrity Summary

Metric	Value
Total Relationships Analyzed	3.0
Critical Integrity Issues	0.0
Total Orphaned Records	903.0

Detailed Integrity Analysis

Critical Issues (3 relationships)

Source → Target	Key Field	Integrity %	Orphaned Records	Business Impact
Activities.ClassCode → Activities.Activity Code (report-only)	ClassCode→Activity Code	0.0%	451.0	Medium: 451/451 records with integrity issues - impacts data quality
Activities.ClassCode → Activity Links.Child ActivityCode (report-only)	ClassCode→Child ActivityCode	0.0%	451.0	Medium: 451/451 activity references invalid - impacts reporting completeness
Activities.SessionParts.ClassCode → Activity Links.Child ActivityCode (report-only)	ClassCode→Child ActivityCode	0.0%	1.0	Medium: 1/1 activity references invalid - impacts reporting completeness

Cross-File Error Patterns (Legacy Analysis)

No cross-file error patterns detected.